Tactix: DSR checker-route gate and WV100 launch

Codex for Andrew

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Objective and outcome. The immediate objective was to make the newly approved native-DSR verification contract operational, then submit the full 100-instance winner-vs-Tactix (WV100) Phase 2 gate. This increment is implemented and pushed as source commit `0a906338`; pinned array `1731686_0-99` is running. At the report timestamp, 91/100 final records are independently checked and valid: 30 original-CNF SAT models and 61 complete UNSAT proof routes. Nine rows, including `ncc_none_21015`, remain. The full Phase 2 gate has not been accepted, and no Tier D job has been submitted.

Checker contract. Every native DSR proof now goes first through pinned no-option `dsr-trim`. A witness-free LSR elaboration finishes with CakeLPR; one containing an SR witness finishes with Trestle. The solver writes one `c proof-route` declaration on an UNSAT proof-producing run based on whether its proof writer actually emitted an SR witness. The production runtime follows that declaration and scans the certificate only to reject a mismatch. The independent checkers still see a clause-sequence-preserving spelling of the original CNF. Startup PBP remains VeriPB then CakePB; explicit DRAT/LRAT remain compatibility paths. No legal public X7 action or proof rule was narrowed, and no row-specific checker setting was introduced. This change replaces the earlier witness-free DRAT-trim route because the user explicitly selected the general DSR-trim-first contract after the archived `ncc` diagnostic; the current-source `ncc` certificate is a different proof, so the archived timing cannot certify this campaign.

Correctness evidence. The checker/runtime/fuzz protocol suite passed 113 Python tests; Release and Debug proof serializer tests and the current-source Debug `micro_proofs` suite passed against independent checkers. The previous repaired-source full Debug sweep completed 188/188 but predates this route edit. The accepted pre-route Phase 1 arrays checked Rank 1–50 at 50/50 (25 SAT models, 25 UNSAT proofs, 7,553.134 solver seconds versus a strict 8,115.735-second target) and B28+C5 at 33/33 (17 models, 16 proofs, 905.892 versus 952.263 seconds). Latest-source proofless Rank finished 50/50 with all 25 SAT models checked; proof-producing performance and proof checks under the new route still need current-source confirmation. At 21:56 UTC, the 91 AM507 final records had matching verdicts, complete independent checks and cleanup; 30 were SAT, 61 UNSAT. This is interim evidence only: the pinned scorer requires exactly 100 final records, 33 SAT models and 67 UNSAT proofs. Early UNSAT routes include both CakeLPR and Trestle after `dsr-trim`.

Resource and provenance boundary. The submitted bundle freezes source `0a906338`, Release binary SHA-256 `cab44387ea47f3ed5d74cbf1a9bb702f321cf9164a3df791ef4b07d760c08656`, all 100 input hashes, and seven checker pins. The full manifest digest is recorded in the AM507 launch report. The unthrottled 100-row array requests `cpu-dense-hwulab-q`, 12 CPUs and 32 GiB

per task, a 1,200-second solve window and a 57,600-second checker window without overrides. The older AM500 old-route bundle is frozen and unsubmitted. These limits belong to WV100, not the eventual Tier D run.

Unresolved obligations. The exact two-stage exit definition is proposed in the AM508 acceptance report. Stage A is the finite, independently checked preflight that authorizes the full Tier D submission; Stage B scores that run and decides N3. This avoids requiring Tier D evidence before Tier D can be launched. The proposal is not a waiver or an accepted change to the plan.

First, finish AM507 and investigate any timeout or rejected proof generally. Then reconfirm current-source Phase 1 and full Debug coverage, run fresh Opus-owned default/random-policy fuzzing, establish public reach of the direct-Proof-API D2F retained-deletion failure, and close broad N1/N2 proof/model and rule-action coverage. The CaDiCaL/Kissat clausal emitter map is structurally complete, while the general live pseudo-Boolean arithmetic and redundancy interface used by LymphoSAT/Satsuma remains a documented gap. N3 still needs matched Rank attribution, checked WV2–89 at its 600-second limit, Tier B/C parity and Tier D Kissat/VBS performance. N4 still needs its scheduling parity ledger closed or explicitly accepted. An online-learning readiness review must address adjustable inprocessing dose/order and switchable branching and clause-deletion strategies with correctness and overhead evidence. Only after the agreed gates are verified or explicitly revised should the full 1,166-instance Tier D array be submitted at 5,000 seconds, four CPUs and 32 GiB per instance. Any relaxed gate must be reported as unverified, not passed.

Primary evidence. `benchmarks/reports/2026-09-27-am506-dsr-trim-first-gate.md`; `benchmarks/reports/2026-09-27-am507-wv100-dsr-trim-first-launched.md`; `benchmarks/reports/2026-09-27-am508-goal-exit-contract-proposal.md`; `benchmarks/reports/2026-09-27-am502-repaired-phase1-terminal.md`; `benchmarks/reports/2026-09-27-am505-proofless-rank-terminal.md`; `benchmarks/reports/2026-09-27-am493-proof-rule-coverage-current-pin.md`; `benchmarks/reports/2026-09-25-am391-learning-readiness-current-source.md`.